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Mechanistic insights into the lignin dissolution behavior in amino acid based deep eutectic solvents

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ABSTRACT

Deep eutectic solvents (DESs) composed by amino acids (L-arginine, L-proline, L-alanine) as the hydrogen bond acceptors (HBAs) and carboxylic acids (formic acid, acetic acid, lactic acid, levulinic acid) as hydrogen bond donors (HBDs) were prepared and used for the dissolution of dealkaline lignin (DAL). The mechanism of lignin dissolution in DESs was explored at molecular level by combining the analysis of Kamlet-Taft (K-T) solvatochromic parameters, FTIR spectrum and density functional theory (DFT) calculations of DESs. Firstly, it was found that the formation of new hydrogen bonds between lignin and DESs mainly drove the dissolution of lignin, which were accompanied by the erosion of hydrogen bond networks in both lignin and DESs. The nature of hydrogen bond network within DESs was fundamentally determined by the type and number of functional groups in both HBA and HBD, which affected its ability to form hydrogen bond with lignin. One hydroxyl group and carboxyl group in HBDs provided active protons, which facilitated proton-catalyzed cleavage of β-O-4, thus enhancing the dissolution of DESs. The superfluous functional group resulted in more extensive and stronger hydrogen bond network in the DESs, thus decreasing the lignin dissolving ability. Moreover, it was found that lignin solubility had a closed positive correlation with the subtraction value of α and β (net hydrogen donating ability) of DESs. Among all the investigated DESs, L-alanine/formic acid (1:3) with the strong hydrogen-bond donating ability (acidity), weak hydrogen-bond accepting ability (basicity) and small steric-hindrance effect showed the best lignin dissolving ability (23.99 wt%, 60 °C). On top of that, the value of α and β of L-proline/carboxylic acids DESs showed some positive correlation with the global electrostatic potential (ESP) maxima and minima of the corresponding DESs respectively, indicating the analysis of ESP quantitative distributions of DESs could be an effective tool for DESs screening and design for lignin dissolution as well as other applications.

1. Introduction

Lignin is the second-most abundant natural biopolymer on the earth, which has received growing attentions in the field of alternative sustainable materials worldwide at both academic and industrial levels in the past decades, due to the threat of irreversible exhaustion of global energy resources and increasingly serious environmental pollution problems [1,2]. Considering the unique properties of lignin including biodegradability, biocompatibility, low cytotoxicity, oxidation resistance and anti-ultraviolet radiation, numerous researches have been conducted to transfer lignin to various value-added products such as lignin-based spherical particles, carbon nanotube fibers, phenolic resin, biofuels, rigid polyurethane foams, epoxies and polyester, enabling lignin with broad applications including adsorbent, drug delivery, adhesive, cosmetics, sustainable construction and packaging [3–8]. In spite of the interesting properties and great potential application, the large-scale application of lignin has not been achieved so for, greatly inferior to that of hemicellulose and cellulose. This is caused by

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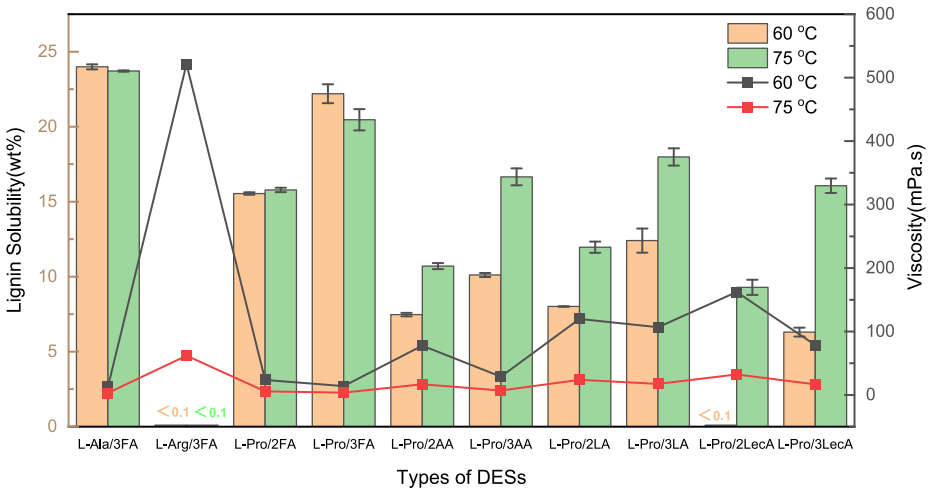


Fig. 1. The DAL solubility (column) vs viscosity (line) of various DESs at 60 and 75 °C.

the heterogeneous and intractable structure of lignin, which is composed of the random cross link of three phenylpropanoid monomers through various interunit linkages mainly including β-O-4, β-β, β-5, and 5-5′, making the effective and green dissolution of lignin still a challenge, which in turn cause low reactivity and marketing value of lignin [9]. Kraft lignin, one major industrial lignin with high annual output of 55 M tons obtained from Kraft pulp, is mainly used for heat recovery [10]. In this study, one commercial type of kraft lignin, DAL was chosen to evaluate the lignin dissolving ability of various DESs.

Organic-based solvents are important and traditional mediums used for lignin dissolution, mainly including alcohol, carboxylic acid, heterocyclic compound and their corresponding aqueous solution [11]. However, the organic-based solvents are normally associated with some drawbacks such as equipment corrosion, high volatility, toxicity and flammability. In the past few decades, ionic liquids (ILs) have attracted considerable attention, due to the tunable physicochemical property, but still possess some inherent shortcomings including complex synthesis, expensive raw materials, questionable toxicity and poor recoverability [12,13]. Deep eutectic solvents (DESs) have been considered as more economical and environmentally friendly alternative solvents to conventional ILs, due to some significant merits over conventional ILs including low cost, biodegradability, low-toxicity and easy one-pot preparation method, as well as sharing some similar properties to ILs such as task-specific character, large liquid temperature ranges, high ionic conductivities, wide electrochemical window, negligible vapor pressure and low flammability [14,15]. Most noteworthy point is that changing the components or molar ratio of hydrogen bond accepter (HBA) and hydrogen bond donator (HBD) in DESs can lead to the tunable nature in physiochemical properties, making it potential to be used as task-specific DES due to the numerous combinations of DESs [16]. Except as dissolving medium, DESs can act as reaction media or catalysts for lignin valorization into valuable aromatic compounds through several reactions such as depolymerization, nano-miniaturization and esterification [17–21]. Apart from that, DESs can be used to make the biomass be selectively fractionated to cellulose-rich or lignin-rich fractions, as different DESs showed different dissolving abilities for lignin and cellulose [21]. Carboxylic acid based DESs have been reported to be effective on dissolving lignin, due to the higher degree of protonation (higher ionicity) [22,23]. From the reported literatures, it was concluded that molar ratio of HBD and HBA, as well as the functional group and carbon chain length of HBD and HBA showed a vital role in lignin dissolving ability of DESs [23–29]. Malaeke et al. [27] found that both hydroxyl and phenyl group played an important role through the order of the dissolving ability of four ChCl DESs with different HBDs: resorcinol > phenol > maleic acid > α-naphthol, also indicating that two hydroxyl groups (resorcinol) was desirable but two phenyl groups (α-naphthol) was not conductive to dissolve lignin. Liu et al. [30] found the enhancement of different functional groups in HBA on the lignin solubility in LA based DESs was in the order: C=C > -OH > benzene ring, by using four quaternary ammonium salts as HBA. However, the conclusions were experiential and the reasons were speculative and needed to be validated. Additional in-depth investigations on the effect of different functional groups on the lignin dissolving ability of DESs at molecular level were required.

Currently, the previous researches mostly focused on ChCl or other quaternary ammonium salts, it is necessary to explore a broader range of DESs used for lignin dissolution and lignocellulose pretreatment. Amino acid was also a good choice as HBD, due to environmental benign, biodegradability and low toxicity [31]. Amino acids based DESs have been applied in many research fields such as solubility of drugs, polymer production, depolymerization of lignocellulose and extraction of bioactive compounds [32–35]. However, there have been very few reports pertaining to the amino acids based DESs used for lignin dissolution. The coexistence of amino and carboxyl groups in amino acids endowed amino acids based DESs tremendous potential as dissolving media, because both amino and carboxyl groups could form hydrogen bond with solutes [36]. Therefore, L-alanine (L-Ala), L-arginine (L-Arg) and L-proline (L-Pro) were selected as HBA to formulate DESs with various carboxylic acid including formic acid (FA), acetic acid (AA), lactic acid (LA) and levulinic acid (LevA) to screen out the suit DESs for lignin dissolution in this study. These carboxylic acids were selected as HBD not only because their corresponding amino acids DESs had not been investigated before, in addition, to investigate the effect of alkyl chain length, hydroxyl, carbonyl and carboxyl groups on the properties and lignin dissolving ability of DESs. Despite the numerous data of lignin solubility on various DESs, little has been made to elaborate the synergistic effects of different functional groups in DESs as well as the lignin solubilization mechanism from the view of molecular interactions [9,37]. We hypothesized that the number and type of functional group in HBD and HBA affected the hydrogen bond network within DESs, which further determined the ability of DESs to interact with lignin. To prove this hypothesis, the relationship between lignin solubility in various DESs and their nature of hydrogen bond network was explored by DFT calculations, K-T parameters and FTIR.

2. Experimental section

2.1. Materials

DAL was purchased from Tokyo Chemical Industry (TCI, Shanghai).

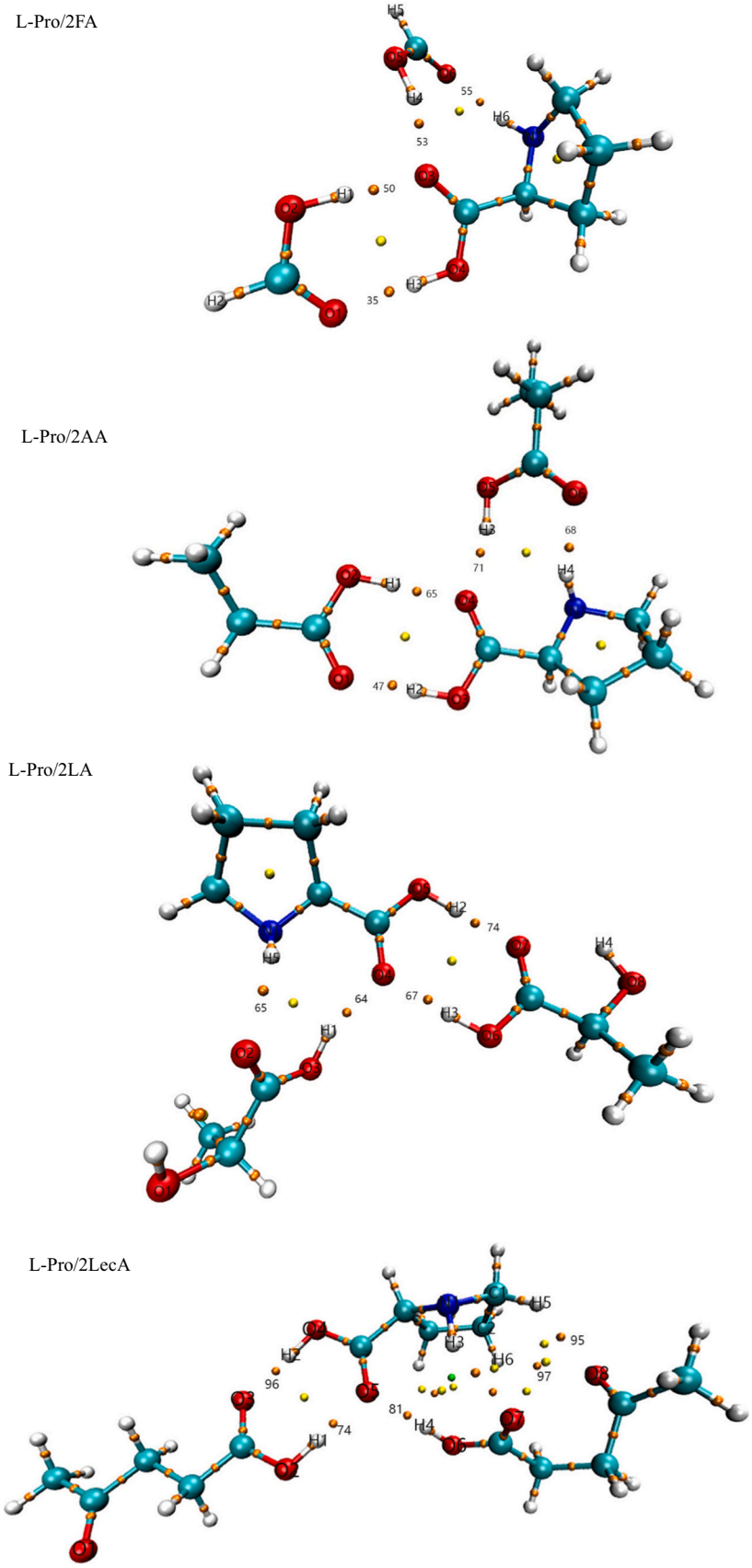


Fig. 2. AIM molecular graphs of L-Pro/2FA, L-Pro/2AA, L-Pro/2LA and L-Pro/2LecA.

Table 1
ρ_b, $\nabla^2\rho$ and hydrogen bond energy at BCP of various DESs.

DES	BCP	Bond	ρ(BCP)/a.u.	$\nabla^2\rho$(BCP)/a.u.	Bond energy (kcal/mol)	Strength
L-Pro/2FA	35	O4-H3…O1	0.045	0.132	−9.22	Weak to medium
	50	O2-H1…O3	0.040	0.125	−8.22	Weak to medium
	53	O5-H4…O3	0.032	0.118	−6.48	Weak to medium
	55	N1-H6…O6	0.016	0.056	−2.87	Weak to medium
L-Pro/2AA	47	O3-H2…O1	0.048	0.147	−10.07	Weak to medium
	65	O2-H1…O4	0.038	0.133	−7.73	Weak to medium
	71	O5-H3…O4	0.030	0.122	−5.92	Weak to medium
	68	N1-H4…O6	0.019	0.072	−3.58	Weak to medium
L-Pro/2LecA	96	O4-H2…O3	0.048	0.147	−9.86	Weak to medium
	74	O2-H1…O5	0.040	0.137	−8.08	Weak to medium
	81	O6-H4…O5	0.032	0.123	−6.42	Weak to medium
	82	N1-H3…O7	0.017	0.061	−2.96	Weak to medium
	95	C1-H5…O8	0.006	0.021	−0.65	Weak
	97	C2-H6…O8	0.008	0.028	−1.12	Weak
L-Pro/2LA	74	O5-H2…O7	0.045	0.144	−9.23	Weak to medium
	67	O6-H3…O4	0.041	0.139	−8.33	Weak to medium
	64	O3-H1…O4	0.029	0.115	−5.72	Weak to medium
	65	N1-H5…O2	0.018	0.067	−3.25	Weak to medium

L-arginine (L-Arg) (98 %) and DL-lactic acid (LA) (85–90 %) were obtained from Rhawn Bio-Chem Technology Co. (Shanghai), L-proline (L-Pro) (99 %), L-alanine (L-Ala) (99 %) was purchased from Macklin Biochemical Co. (Shanghai), acetic acid (AA) (99.5 %) was purchased form Yongda Chemical Reagent Co. (Shanghai), Formic acid (FA) (99 %) and levulinic acid (LevA) (99 %) was purchased from Aladdin Bio-Chem Technology Co. (Shanghai), 4-nitroaniline (99 %) and Nile red (97.5 %) were obtained from J&K Scientific Co. (Guangdong). All the above chemicals and reagents were used of analytical grade without further purification.

2.2. DES preparation

The preparation of DESs was carried out by one-step heating method, which was based on mixing the HBA and HBD at molar ratio of 1:2 and 1:3 followed by heating at 100 °C in oil bath under constant stirring until a homogeneous liquid was formed. The obtained transparent liquid was cooled at room temperature, transferred to a vacuum drying oven, kept under vacuum at 60 °C for 24 h to remove the contaminated trace water [38]. Then the water content was measured by using a ZSD-2J Karl Fisher titrator. The water content of all the DESs were lower than 2 wt% (Table S2), which was far to affect the supramolecular structure of DESs [39]. The prepared DESs were denoted as HBD/xHBA, where x represented the molar ratio of HBD and HBA.

2.3. Lignin solubility assays

All the prepared DESs were used to dissolve lignin. The lignin solubility was determined by cloud point method, which has been used in most reports [27,28,40,41]. The DES (ca. 2.5 g) was added to a round-bottom flask with a magnetic stirring bar, and then was transferred to oil bath at some specific temperature. Continue adding lignin (about 20 mg at each time) when the solution is optically transparent under the microscope until the added lignin did not dissolve within 4 h. One drop of the DESs was placed in the centre of the slide and covered by micro coverslips slowly. Then it was observed under light microscope at the magnification of 10 × 10. The cumulative addition amount of lignin (m_{lignin}) which did not include the last addition amount of lignin, was used to determine solubility. For the DESs that cannot dissolve 20 mg lignin in 4 h, firstly add 2 mg lignin to 2.5 g of DES to determine if it can dissolve lignin. The solubility of lignin in DESs was calculated by the following equation, where $Solubility_{lignin}$ (wt%) refers to the solubility of lignin in DESs, m_{lignin} refers to the cumulative addition amount of lignin, m_{DES} refers to the mass of DES used for dissolution. All solubility tests were repeated twice and the results presented were the average of twice measurements. The standard deviations were given in the supplemental materials, except the cases with a solubility of <0.1 wt%, because of the too low solubility.

$$Solubility_{lignin} = \frac{m_{lignin}}{m_{lignin} + m_{DES}} \times 100\%.$$

2.4. Regeneration of lignin in DESs

The dissolved lignin in L-Pro/3FA at 60 °C was regenerated by using deionized water (20 ml) as antisolvent at room temperature with vigorous stirring for 30 min and then centrifuged. The obtained lignin was washed by deionized water and centrifuged by several times to ensure complete removal of DES, and finally dried in mild oven conditions (60 °C).

2.5. DESs recovery and recycling

The supernatant liquid obtained from the centrifugation of the mixture of L-Pro/3FA/lignin and water, was transferred to the vacuo rotary evaporator and kept at 60 °C for 2 h, which was adequate to remove almost all the water from the L-Pro/3FA solvent system [42]. Then the recovered DESs was used to dissolve DAL at 60 °C to evaluate the DESs recycling performance (5 runs). The DAL solubility in L-Pro/3FA was used to evaluate the DESs recycling performance. All assays were carried out twice.

2.6. K–T solvatochromic parameters measurement

The dyes including 4-nitroaniline (NH2) and Nile red (NR) were used to measure K-T solvatochromic parameters. The dyes were firstly dissolved in methanol (1.0×10^{-3} mol/L) and stored in amber glass vials. Each time 50 μL dye solution was added to the centrifuge tube. Methanol was evaporated by vacuum drying at 40 °C for 0.5 h. The dried dye was mixed with 2 g DES and transferred to quartz cells with 1.0 cm light path length for examination. UV visible spectrophotometer was used to record the absorption spectrum of the DES-dye mix in the wavelength range of 300 to 700 nm at room temperature (25 °C). α, β and π were calculated using the following formula [43], $\lambda_{(NR)max}$ and $\lambda_{(NH2)max}$ represented the maximum absorption wavelength of Nile red and 4-nitroaniline:

$$\alpha = \frac{19.9675 - 1.0241 \times \pi - \nu_{(NR)max}}{1.6078}$$

$$\beta = 11.134 - \frac{3580}{\lambda_{(NH2)max}} - 1.125^{*}\pi$$

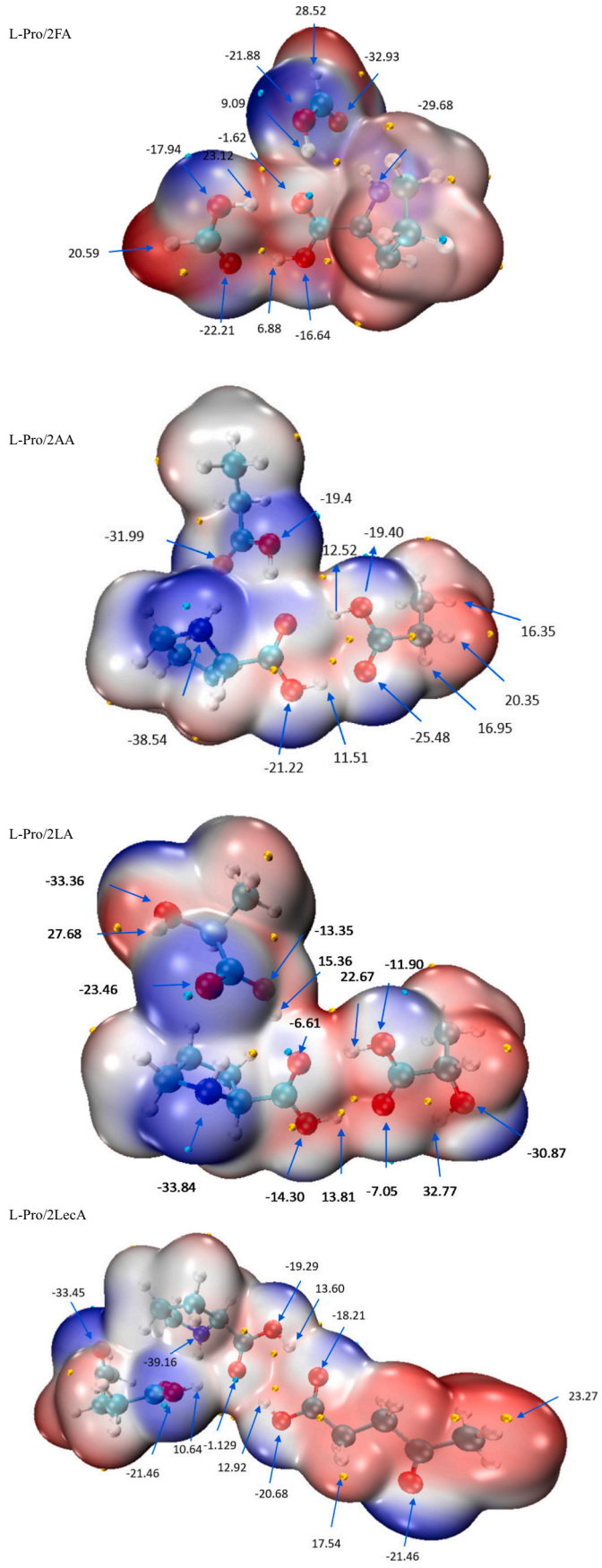


Fig. 3. ESP mapped on electron total density with an isovalue 0.001 of the structurally optimized L-Pro/2FA, L-Pro/2LA, L-Pro/2AA and L-Pro/2LecA.

$$\pi = \frac{19.839 - \nu_{NR}}{2.9912}$$

$$\nu_{NR} = 10000/\lambda_{(NR)max}$$

2.7. Viscosity measurement of DESs

The viscosity of DESs was measured by the Pinkevitch method [44] and the viscometer was calibrated by high purity glycerol in a super constant temperature bath with a temperature control accuracy of ±0.01 K. The density of DESs was detected by the Pycnometer method (±0.0005 g·cm^{-3}) and the pycnometer was calibrated with deionized water. The viscosity and density of DESs were calculated by the following equations:

2.8. FTIR analysis of DESs and DAL

The FTIR spectrums were recorded using KBr plates on a FTIR spectrometer (Bruker Tensor 27, Germany) with wavenumbers from 4000 to 400 cm^{-1} at a resolution of 4 cm^{-1}. The DAL samples were characterized with a thin KBr disk prepared under high pressure, and liquid DESs samples were prepared by dripping liquid on the blank KBr disk.

2.9. ^{1}HNMR analysis of DESs

A 300 μL aliquot of DESs was thoroughly mixed with 300 μL deuterated water, then transferred to an NMR tube. ^{1}H NMR was recorded by using a nuclear magnetic resonance spectrometer (Bruker, Avance 500 MHz, Germany) at room temperature. Relaxation delay was set to 5. Approximately 1000 transients were acquired to ensure a high signal/noise ratio [45].

2.10. Thermogravimetry analysis (TGA) of DAL

The TGA analysis were performed using a NETZSCH STA449F3 (NETZSCH Scientific Instruments, Germany) under N_2 atmosphere at the temperature range of 50–700 °C with the heating rate of 10 °C/min.

2.11. Scanning electron microscopy analysis (SEM) of DAL

The SEM measurements were performed on a Hitachi S-4800 Scanning Electron Microscope operated at 5 kV. The samples were spray-coated with a thin layer of platinum before observation.

2.12. Quantum chemical calculation

For geometry optimization of DESs and the followed electrostatic potential (ESP) analysis, Atoms in molecules technique (AIM) analysis and Reduced Density Gradient (RDG) analysis, Gaussian09, Molclus, Multiwfn and VMD 1.9.3 software programs were used for related calculation. The starting DES cluster configurations were generated by the genmer program under Molclus software which was a very easy-to-use program for searching cluster configurations [46] and then pre-optimized at pm6D3 through Gaussian09 [47], which was cheap and fast. Then the selected low-energy DESs structures were reoptimized at the higher M06-2X/6-311++G(d,p) level through Gaussian09, which has been proved to be high accurate for geometry optimization and vibrational analysis of DESs [48]. The energy minima of all the optimized DESs were confirmed by the absence of imaginary frequencies. The ESP colored molecular surface map with ESP surface extrema, the AIM topology map and the RDG map of DESs was plotted by VMD 1.9.1 program [49] based on the output from Multiwfn 3.8 [50] which was a multifunctional wavefunction analysis program. Simultaneously, the various properties including the electron density (ρ_b), the Laplacian of

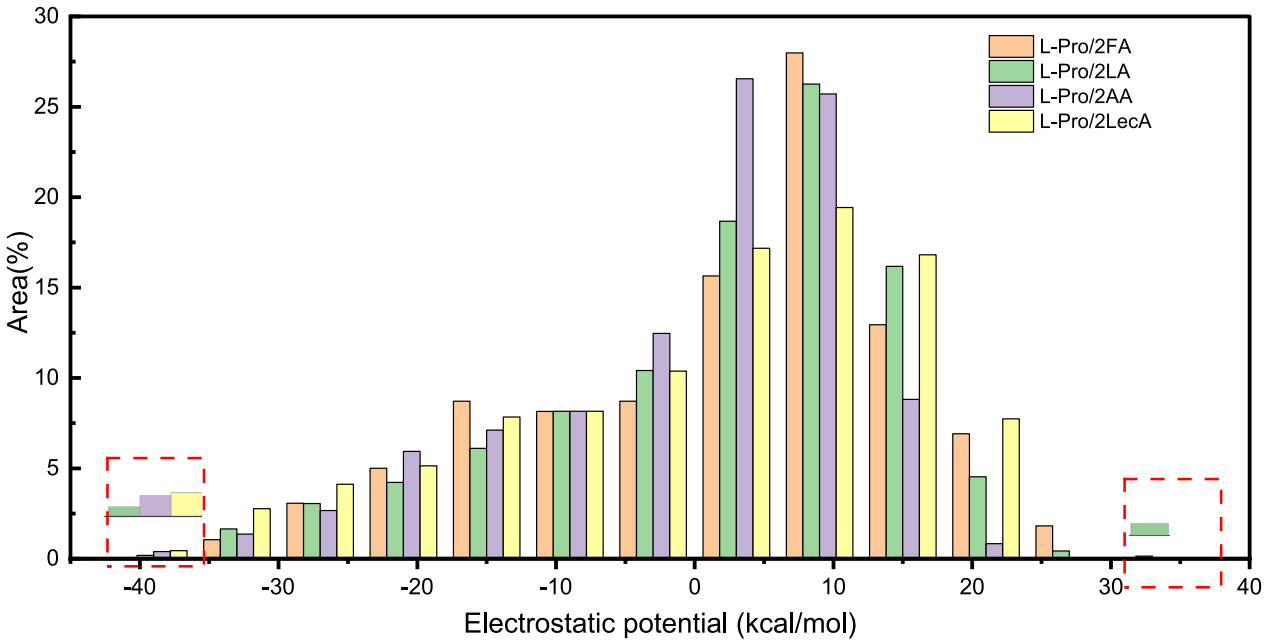


Fig. 4. The electrostatic potential quantitative distributions of the structurally optimized L-Pro/2FA, L-Pro/2LA, L-Pro/2AA and L-Pro/2LecA.

Table 2
K-T empirical parameters of L-Pro, L-Ala and L-Arg DESs.

DES	Molar ratio	α	β	π	α-β
L-Pro/FA	1:2	1.23	0.56	0.94	0.67
L-Pro/FA	1:3	1.28	0.35	0.99	0.93
L-Pro/AA	1:2	0.84	1.14	0.62	−0.30
L-Pro/AA	1:3	0.98	0.86	0.74	0.12
L-Pro/LA	1:2	1.03	0.96	0.78	0.07
L-Pro/LA	1:3	1.09	0.81	0.83	0.29
L-Pro/LecA	1:2	0.84	1.40	0.62	−0.56
L-Pro/LecA	1:3	1.14	1.05	0.87	0.09
L-Ala/FA	1:3	1.33	0.33	1.02	1.00
L-Arg/LA	1:3	1.23	0.68	0.94	0.54

the electron density ($\nabla^2\rho_b$) at the bond critical point (BCP) were obtained from the output of Multiwfn 3.8.

3. Results and discussions

Three amino acids namely L-Ala, L-Arg and L-Pro were selected as HBD to form DESs in this study. Compared to L-Arg and L-Pro, it was more difficulty to prepare L-Ala based DESs, ascribed to the strong interaction energy between L-Ala molecules. In this work, L-Ala could only form DES with FA at molar ratio of 1:3. L-Arg based DESs could be easily prepared, but possessed high viscosity. Therefore, L-Pro was used to prepare DESs with various carboxylic acids to give a suitable selection of HBD and explore the effect of molar ratio of HBD to HBA. L-Ala/3FA and L-Arg/3FA were prepared to elaborate the effect of the structure of HBA. All the prepared DESs were listed in Table S1.

3.1. Effect of the molar ratio and temperature

All the prepared amino acid based DESs have been tested the lignin dissolving ability at 60 and 75 °C, presented in Fig. 1. The numerical values of solubilities and corresponding standard deviations were provided in Table S3. Fig. 1 showed that the DAL solubility in all L-Pro based DESs was significantly enhanced when the molar ratio of HBD to HBA increased from 1:2 to 1:3, consistent with the reported experiments in which ChCl was used as HBA [23,28,51]. L-Pro/3FA showed the DAL solubility of 22.20 wt% at 60 °C, which was 1.42 folds to that of L-Pro/2FA. Analogously, the increase of molar ratio from 1:2 to 1:3 in L-Pro/LA, L-Pro/AA and L-Pro/LecA enabled an increase in DAL solubility of 4.39, 2.64 and 6.31 wt%, respectively. This was because the higher content of HBD in DESs contributed to decreasing the viscosity of DESs and providing more active protons, thus allowing more opportunity for interacting with lignin particles [52]. The viscosity, one macroscopic physicochemical property of DESs, is the basic parameter needed to be considered for their suitability. The viscosity of DESs had higher affinities to the number and strength of hydrogen bonds, which was determined by the components, molar ratio and temperature [39]. As illustrated in Table S4, the viscosity of L-Pro based DESs all decreased as the molar ratio increased from 1:2 to 1:3.

The viscosity of all the investigated DESs decreased distinctly as the temperature increased from 60 to 75 °C (Fig. 1), because the intermolecular interaction such as hydrogen bond was weakened and the average distance between molecules was increased [53]. Therefore, the active sites for interacting with lignin became less blocked as temperature increased. Except L-Ala/3FA, L-Pro/2FA and L-Pro/3FA, L-Arg/3FA, the lignin solubility in other DESs obviously increased as temperature was increased from 60 to 75 °C. This was because the viscosity of L-Ala/3FA, L-Pro/2FA and L-Pro/3FA at 60 °C was already very low (10–25 mPa.s). Inversely, the viscosity of L-Arg/3FA at 75 °C was still up to 62 mPa.s. Consequently, DAL was practically insoluble in L-Arg/3FA at both 60 and 75 °C.

3.2. Effect of the structure of HBA

Lignin was insoluble in L-Arg/3FA at 60 and 75 °C, whereas both L-Pro/3FA and L-Ala/3FA showed much stronger lignin dissolving ability. From the chemical structural point of view, the main difference among the three amino acids HBA was that there was one more guanidine group in L-Arg except one amino group and one carboxyl group, compared to L-Pro and L-Ala. The existence of one more guanidine group in L-Arg leaded to a more complex hydrogen network and extensive dimer chains within DESs, which resulted in the much higher viscosity of DESs and interfered with DESs to contact with lignin particles [54]. The viscosity of L-Arg/3FA was around 40 folds to that of L-Pro/3FA and L-Pro/3AA (Table S3). L-Ala/3FA slightly surpassed L-Pro/3FA on dissolving lignin and gave the highest DAL solubility (23.99 wt%) indicating the shorter

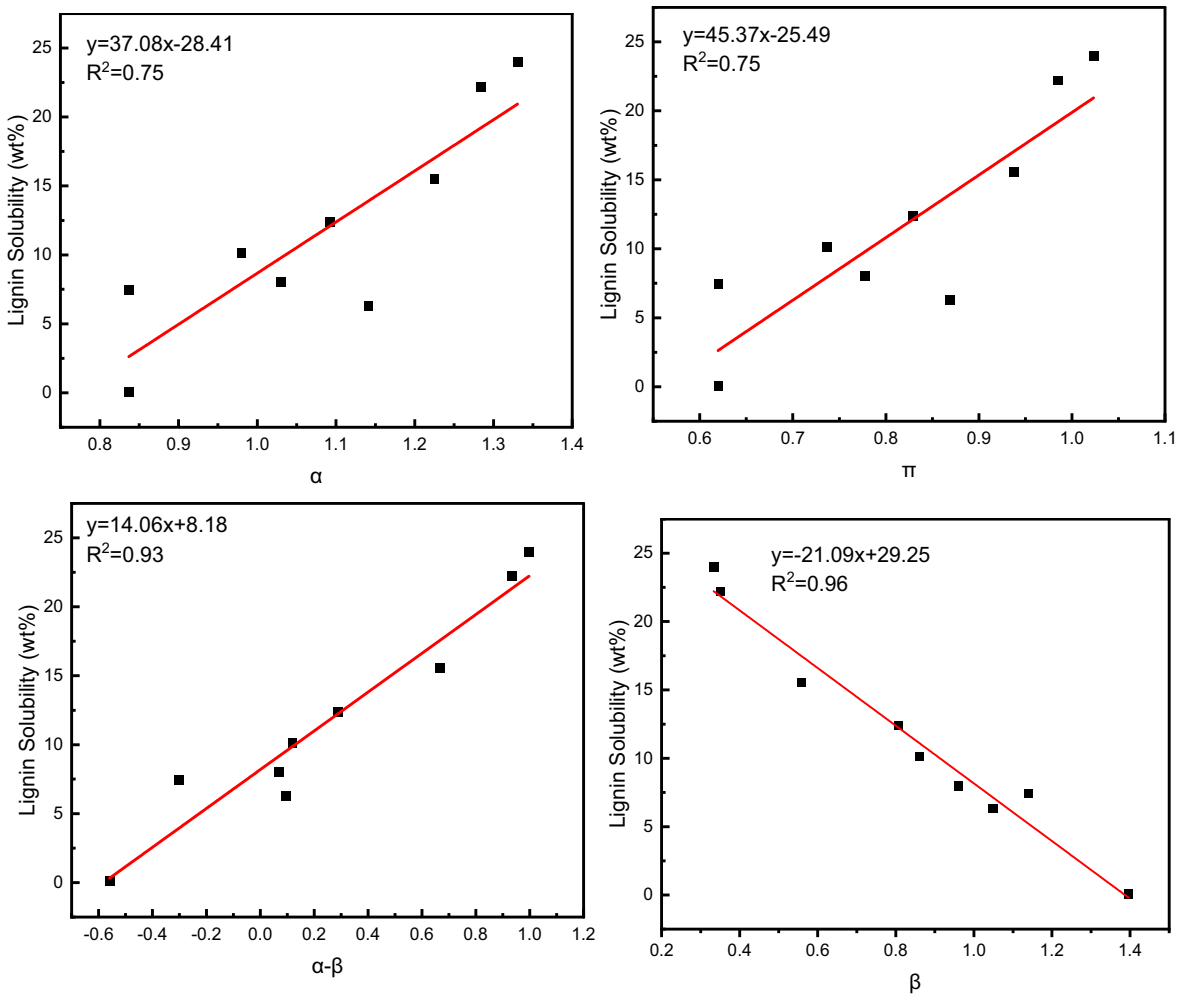


Fig. 5. Relationship between lignin solubility (60 °C) and K-T empirical parameters of L-Pro and L-Ala based DESs.

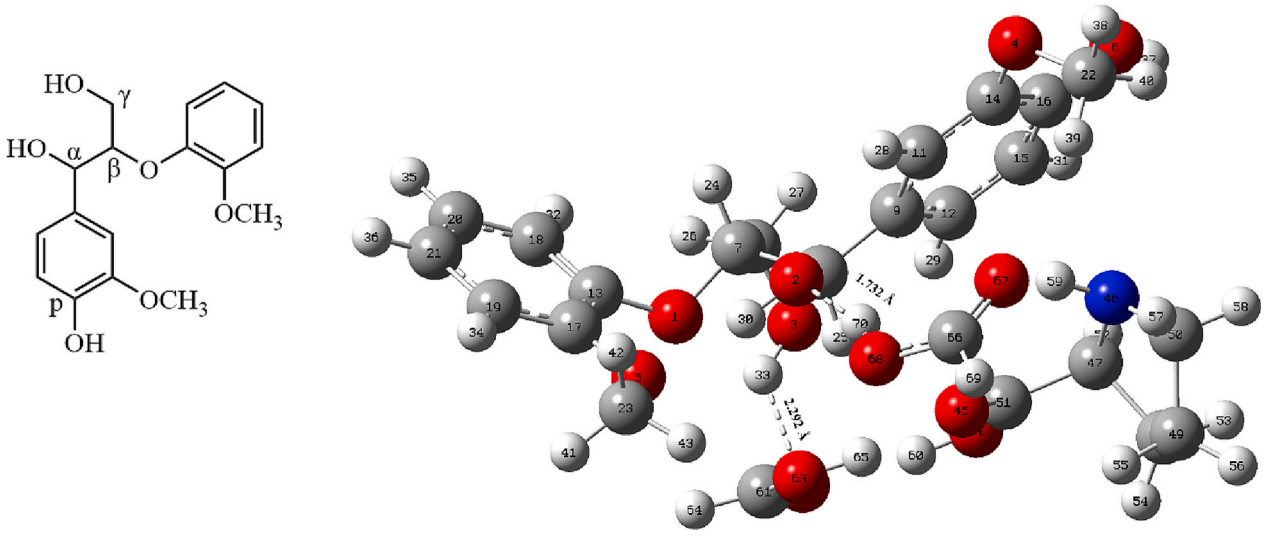


Fig. 6. Chemical Structure of lignin model compound (left), the optimized structure of L-Pro/2FA + lignin model at the M062X/6-311++G** level (right).

alkyl chain length of HBA was favorable for lignin dissolution. Considering the fact that the source and extraction method of lignin also played an important role on the lignin dissolution [27], it is unscientific to directly compare the values of lignin solubility of this study to those from the published literatures, as most literatures did not give detailed introduction of the lignin they used. Therefore, we prepared ChCl/3LA as the controlled group, which was considered to be effective and mostly used for lignin dissolution and extraction. ChCl/3LA gave DAL solubility

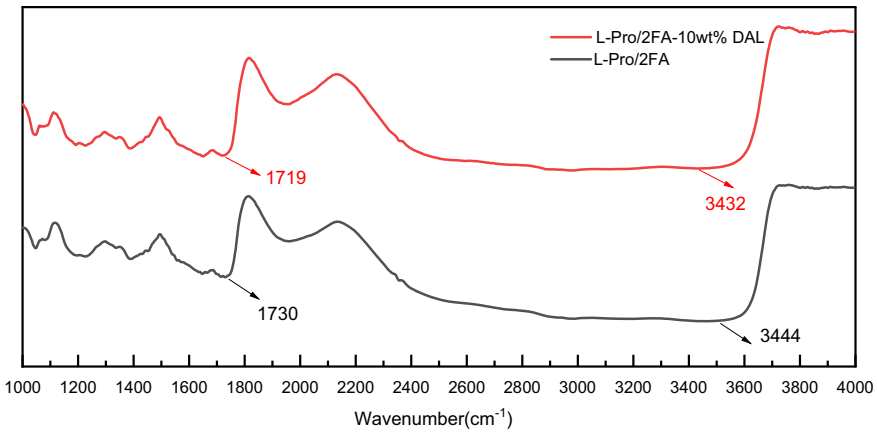


Fig. 7. FTIR spectra of L-Pro/2FA and L-Pro/2FA-10 wt% lignin.

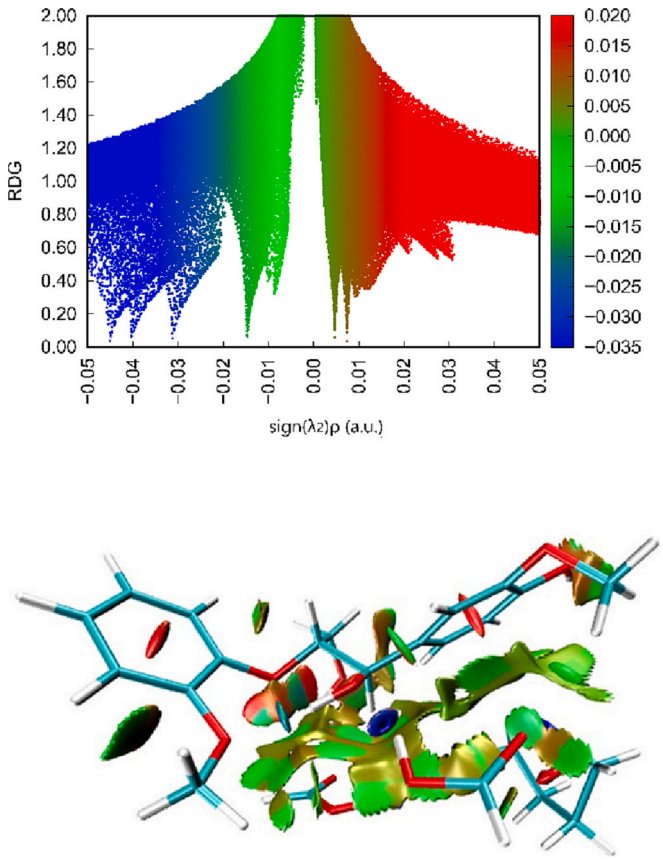


Fig. 8. The RDG scatter plot (up) and colored sign (λ_2) ρ mapped RDG isosurfaces (down) of L-Pro/2FA + lignin.

14.13 wt% at 60 °C in this study, lower than those of L-Ala/3FA (23.99 wt%) and L-Pro/3FA (22.20 wt%), and slightly higher than that of L-Pro/3LA (12.41 wt%), indicating that L-Pro and L-Ala based DESs were potential to dissolve lignin.

3.3. Effect of the structure of HBD

For the L-Pro based DESs at the same molar ratio, FA based DESs performed best, LA based DESs came second, followed by AA, and LecA based DESs (Fig. 1). It was clear that the longer carbon chain and the superfluous functional group in HBD resulted in higher viscosity which was also a measurement of the interaction within DESs to some extent [55], thus hindering the dissolution of lignin. L-Pro/LA was one exception, as LA possessed a longer alkyl chain compared to AA and L-Pro/LA possessed higher viscosity to L-Pro/AA, but L-Pro/LA performed better than L-Pro/AA on dissolving lignin. The superiority of L-Pro/LA may be caused by active protons supplied by additional OH group in LA, which enhanced the ability of L-Pro/LA to interact with lignin. As LecA owned the longest chain length and an addition C═O group, L-Pro/2LecA showed the highest viscosity and poorest lignin dissolving ability. Strangely, L-Pro/3LecA was less viscous than L-Pro/3LA, but performed worse than L-Pro/3LA, indicating that C═O group in LecA might play a negative role in lignin dissolution.

To explore the effect of different functional groups in HBA and HBD on the lignin dissolving ability of DESs more deeply, the atoms in molecules (AIM) theory was used to quantify the hydrogen bond interactions between HBD and HBA in the optimized structure of L-Pro/2FA, L-Pro/2AA, L-Pro/2LA, and L-Pro/2LecA by comparing the electron density (ρ_b) and the Laplacian of the electron density ($\nabla^2\rho_b$) at the bond critical point (BCP) of hydrogen bond [56]. It was crucial that the value of ρ_b of the BCP of a hydrogen bond generally lies within a range of 0.002–0.035 au and was directly proportionate to the strength of hydrogen bond [57,58]. The values of $\nabla^2\rho_b$ of the BCP of non-covalent interactions such as hydrogen bonds generally fall within a range of 0.024–0.139 au [57]. The corresponding hydrogen bond energy was calculated by the equation [59], namely E_{HB} (kcal/mol) = $-223.08 \times \rho_b$/a.u. + 0.7423, used for the classification of hydrogen bond. The classification proposed by Emamian et al. [59] argued that weak, weak to medium, medium and strong hydrogen bond interactions are characterized with the bond energy values, which fall within the range of >-2.5, -2.5 to -14.0, -15 to -11 and <-15.0 kcal/mol, respectively. Fig. 2 showed the AIM molecular graphs of L-Pro/2FA, L-Pro/2AA, L-Pro/2LA and L-Pro/2LecA, wherein the orange points represented BCP. The BCP of hydrogen bonds between HBD and HBA were numbered by black. The ρ_b, $\nabla^2\rho$ and hydrogen bond energy at BCP of L-Pro/2FA, L-Pro/2AA, L-Pro/2LA and L-Pro/2LecA were listed in Table 1. It was found that except L-Pro/2LA, the strength and number of the hydrogen bonds in the other three DESs was positively associated with their viscosity, similar to the result of Wang et al. [60] and Fan et al. [61]. It could be reasonably explained by that the optimized structure of DESs we used did not take the interaction between HBD and HBD into consideration, whereas LA itself possessed much stronger viscosity than AA and FA. L-Pro/2LecA possessed two more weak hydrogen bonds than the other three DESs due to the additional C═O group, resulting in the higher viscosity compared to the other three DESs. In addition, the strongest hydrogen bond of L-Pro/2LecA and L-Pro/2AA was a little stronger than those of L-Pro/2LA and L-Pro/2FA, which made it more difficult for them to react with lignin, thus resulting in the poorer lignin solubility compared to L-Pro/2FA and L-Pro/2LA. This case was

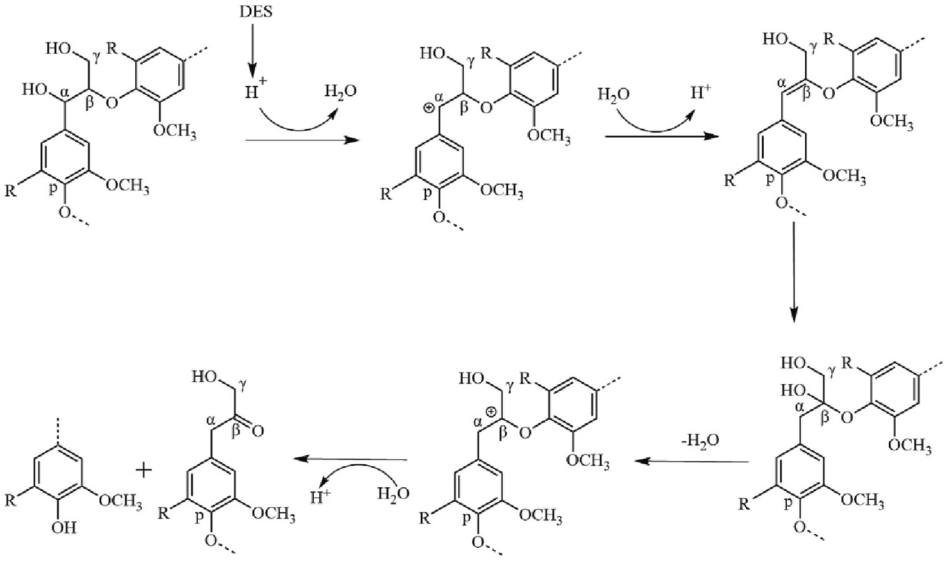


Fig. 9. The possible route of the cleavage of β-O-4′ linkages during lignin dissolution in DESs.

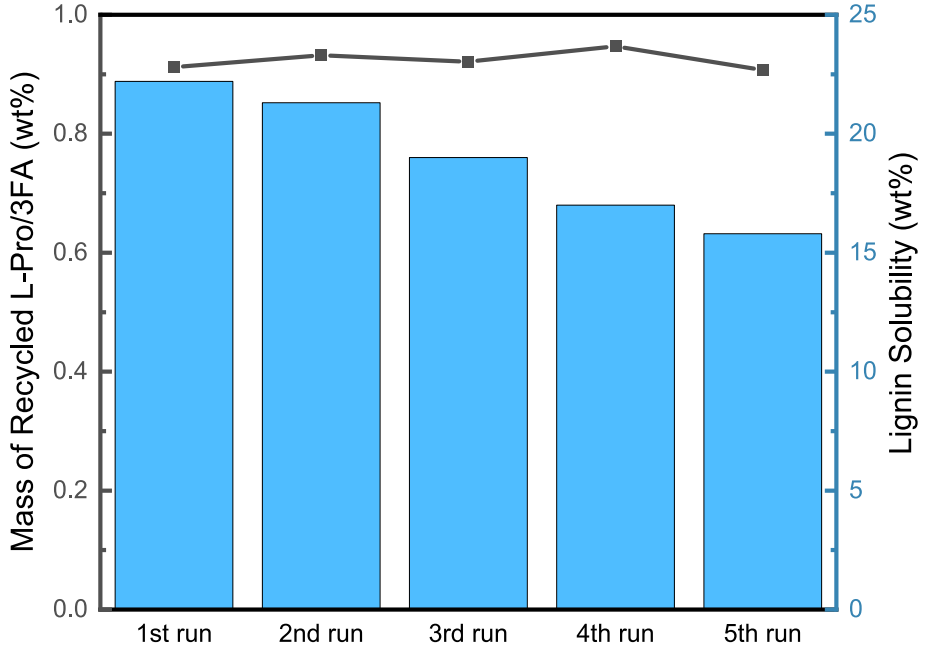


Fig. 10. Recovery rate and DAL solubility in the recycled DESs.

probably because the alkyl group and C═O group donated the electron to the oxygen in the OH group, thus intensifying the intensity of the hydrogen bonds [52]. Therefore, it was concluded that the best performance of L-Pro/2FA among the four DESs might be related with the suit strength of hydrogen bond and small steric hindrance. Inversely, L-Pro/2LecA with the stronger and more hydrogen bonds, performed the worst (0 wt% 60 °C) in dissolving lignin. Our AIM analysis result also supported the speculation of Beaucamp et al. [54] which was proposed based on the summary of representative studies on DESs used for lignin fractionation, that the superfluous functional group resulted in a more extended hydrogen bond network, leading in turn to the weaker ability to interact with lignin.

To support the above assumptions, the electrostatic potential (ESP) quantitative distributions on molecular vdW surface of the structurally optimized L-Pro/2FA, L-Pro/2AA, L-Pro/2LA and L-Pro/2LecA were analysed in Fig. 3, generated based on the published methods [62,63]. As depicted in Fig. 3, the yellow and green sphere corresponded to the electrostatic potential maxima and minima respectively. The strength of the ESP on molecular vdW surface was distinguished by different colors in Fig. 3, wherein blue, red and the remaining colors represented the positive, negative and neutral ESP region respectively. The analysis of ESP quantitative distributions is an effective tool to investigate the intermolecular interaction and predicate the possible reaction sites. It was observed that electronegative area is around O and N atoms while the electropositive area is around H atoms in the four DESs (Fig. 3). The formation of hydrogen bond between HBD and HBA leaded to charge delocalization, which derived the formation of DESs, and simultaneously made the atoms involved in hydrogen bonds possess low electronic density. For example, the electrostatic potential minima generated by O atom in COOH of L-Pro in L-Pro/2FA DES was only −1.62 kcal/mol, as it formed two hydrogen bonds with 2 FA molecules. Similar phenomenon was observed in the other 3 DESs. It was reported

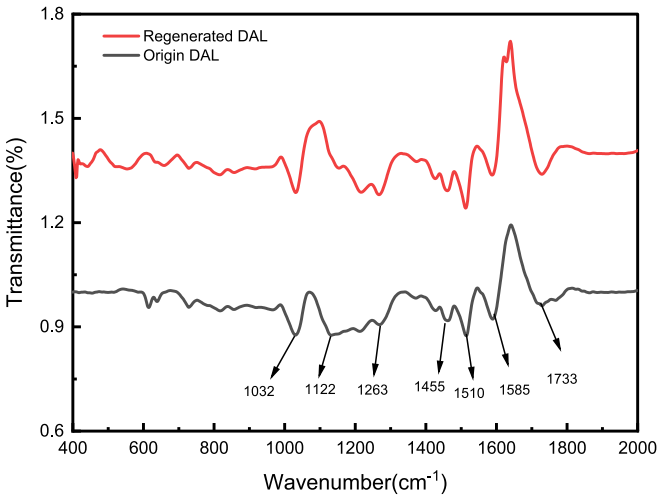


Fig. 11. FTIR spectra of origin and regenerated DAL from L-Pro/3FA.

that the major drive for lignin dissolution was the formation of hydrogen bond between lignin and DESs [64]. The ability of DESs to form hydrogen bonds with lignin was related with the global electrostatic potential maxima and minima of DESs. Fig. 4 provided a more visual and straightforward comparison of the electrostatic potential quantitative distributions of these four DESs, by plotting surface area ratio in different ESP ranges. The ESP on the whole molecular vdW surface was divided into multiple fragments through the quantitative molecular surface analysis module of Multiwfn program, presenting the different electrostatic potential characters of various DESs. The surface area ratio in the ESP ranges of 24 to 30 kcal/mol of L-Pro/2FA was 1.82 %, higher than that (0.43 %) of L-Pro/2LA. In addition, in the ESP ranges of −36 to −30 kcal/mol and −30 to −24 kcal/mol, L-Pro/2LA possessed higher surface area ratio than those of L-Pro/FA. Therefore, L-Pro/2FA possessed stronger hydrogen donating ability and weaker hydrogen accepting ability than L-Pro/LA with high probability. This predication was in good agreement with the experimental K-T empirical parameters (Table 2) of L-Pro/2FA (α:1.23, β:0.56) and L-Pro/2LA (α:1.03, β:0.96). L-Pro/2FA performed much better than L-Pro/2LA in lignin dissolution (15.55 wt% vs 8.01 wt%, 60 °C), indicating that electrostatic potential maxima (hydrogen donating ability) might play a more important role compared to electrostatic potential minima (hydrogen accepting ability) in lignin dissolution in this study. In terms of L-Pro/2AA and L-Pro/2LecA, their global surface electrostatic potential maxima appeared in the ESP rang of 18 to 24 kcal/mol, which was lower than those of L-Pro/2FA and L-Pro/2LA, indicating their weaker hydrogen donating ability, which was in consistent with the result that the α values of L-Pro/2AA (0.84) and L-Pro/2LecA (0.84) were lower than those of L-Pro/2FA (1.23) and L-Pro/2LA (1.03). Conversely, L-Pro/2LecA was considered to possess stronger hydrogen accepting ability than the other three DESs, due to its larger surface area ratio in the ESP range of −36 to −30 kcal/mol and −30 to −24 kcal/mol, caused by its additional C═O. This speculation was also supported by the experimental K-T empirical parameters that the β value of L-Pro/2LecA (1.40) was much higher than those of the three other DESs. Based on the experimental result that lignin solubility in L-Pro/2LecA was <0.1 wt% at 60 °C, it was also implied that the hydrogen accepting ability played a negative role in lignin dissolution in this study.

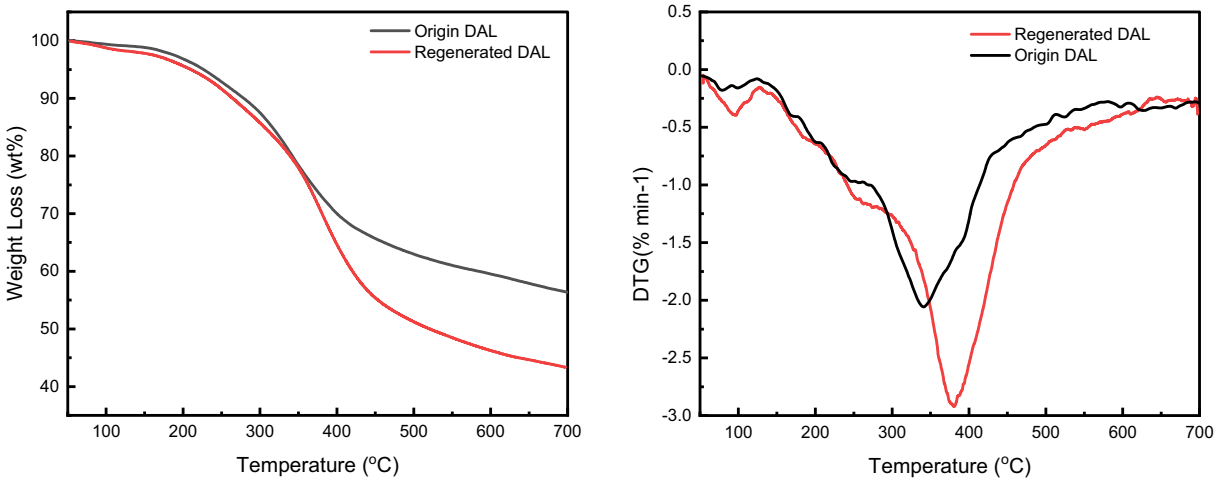


Fig. 12. TGA (left) and DTGA (right) curves of origin DAL and regenerated DAL from L-Pro/3FA DES.

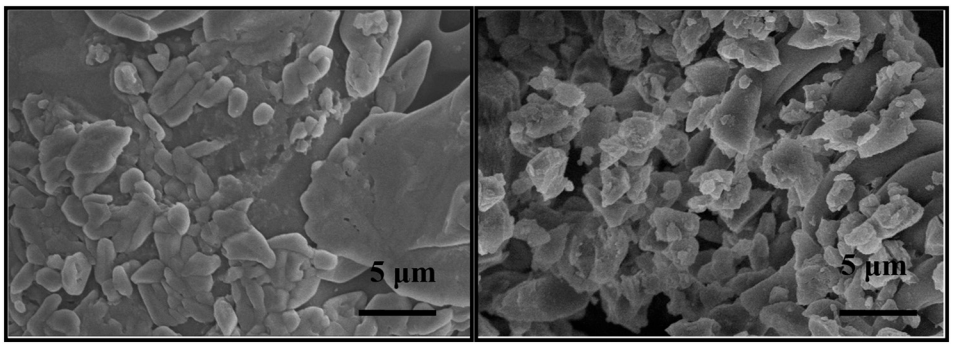


Fig. 13. SEM of origin DAL (left) and regenerated (right) DAL from L-Pro/3FA DES.

3.4. Relationship between K-T empirical parameters of DESs and lignin solubility

The optimized structure of DESs we discussed above must be different from the real structure of DESs, as the interaction between HBD and HBD, HBA and HBA had not been taken into consideration during geometry optimization of DESs. Therefore, K-T solvatochromic parameters of DESs including α, β and π were measured and presented in Table 2, which represented hydrogen-bond donating ability (acidity), hydrogen-bond accepting ability (basicity) and polarity of the real DESs, respectively. The relationship between K-T empirical parameters of DESs and their lignin dissolving ability was expected to be found and further prove the reliability of DFT calculation result above. To provide a throughout and deep understanding of the relationship between K-T empirical parameters of DESs and their lignin dissolving ability, the lignin solubilities (60 °C) versus α, β, α-β and π values of L-Pro and L-Ala based DESs were plotted in Fig. 5. L-Arg based DES was excluded due to the much higher viscosity, which was also a major factor for lignin dissolution [65]. It was found that there were closed positive linear relationships between lignin solubility at 60 °C and α (R^2 0.75), α-β (R^2 0.93) and π (R^2 0.75), and a fairly strong negative relationship between lignin solubility and the value of β (R^2 0.96). Similar results were observed between lignin solubility in various DESs at 75 °C and K-T empirical parameters of DESs, illustrated in Fig. 1S. Based on these findings, it was concluded that the higher net hydrogen donating ability (α-β) was beneficial for lignin dissolution, which was in line with the analysis of ESP quantitative distributions above. Liu et al. [28] and Xue et al. [66] also found that the lignin solubility had a positive correlation with the value of α-β, by using LA/quaternary ammonium salts DESs and ChCl/glycerin, respectively. Therefore, the value of α-β and π of DESs could be an indicator for predicating lignin dissolving ability of DESs. As we discussed in Section 3.3, the larger value and higher surface area ratio of global electrostatic potential maxima and minimum of DESs generally correspond to the larger value of α and β of DESs respectively. Therefore, the analysis of ESP quantitative distributions of DESs is an effective way to screen out DESs for lignin dissolution.

3.5. Mechanism of lignin dissolution in DESs

According to the discussions above, the lignin dissolving ability of DESs was positively correlated with the net hydrogen bond donating ability of DESs. To gain a deeper understanding of lignin solubilization mechanism in DESs, the structure of the mixture (L-Pro/2FA DES/lignin model) was optimized to investigate the interaction between DES and lignin. Monomeric lignin units are randomly linked by various types of C—C bond (~25–35 %) and C—O bond (~60–70 %), of which aryl- and alkyl-ether β-O-4 linkages account about 50 wt% [67,68]. Guaiacylglycerol-β-guaiacyl ether (GG), a typical β-O-4 lignin dimer, which contains both the phenolic and nonphenolic subunits of lignin including β-O-4, aryl group, methoxy group and hydroxy group, have been the most popular phenolic compound as model studies on the dissolution and degradation of lignin [69], as illustrated in Fig. 6 (left). The optimized structure of the mixture (L-Pro/2FA DES/lignin model) was presented in Fig. 6(right), where two new hydrogen bonds formed between GG and L-Pro/2FA DES namely O_{68}-H_{70} (acidic O—H protons of carboxyl group in FA)···O_2 (hydroxy group in C_α of GG) with the bond length of 1.732 Å and O_3-H_{33} (hydroxy group in C_γ of GG)···O_{62} (C═O of carboxyl group in FA) with the bond length of 2.292 Å. Ji and Lv [64] also found that the C–H···π and O···H bonding contributed to the dissolution of lignin in ChCl/LA DES through DFT calculation. It was worth mentioning that DFT calculations always required a high calculation intensity, which limited the size of investigated system and resulted in insuperable gaps between the real DESs system and simplified theoretical models. Therefore, we conduct the FTIR spectra analysis of the system of L-Pro/2FA/lignin. It was observed that the FTIR spectra of L-Pro/2FA/lignin was similar to that of the spectra of the pure DES L-Pro/2FA, which was caused by the same functional groups in DES and lignin as well as the high content of DES in the system, similar with the result of Hladnik et al. [70]. As illustrated in Fig. 7, the C═O stretching vibration and the center of the strong broad OH stretching band in the FTIR spectra of L-Pro/2FA red shifted from 1730 to 1719 cm^{-1} and 3444 to 3432 cm^{-1} after the dissolution of DAL respectively, implying the carbonyl group and hydroxyl group in L-Pro/2FA DES formed new hydrogen bonds with DAL, which was consistent with the DFT calculation result. Xue et al. [66] also discovered the peak of C═O stretching in the FTIR spectra of lignin blue shifted after dissolving in DES. In addition, the melting peak of the mixture of the DES and lignin disappeared compared to that of pure DESs, also suggesting lignin interacted favourably with lignin.

Reduced density gradient (RDG) analysis was applied to further identify the type and strength of non-covalent interaction, by plotting the RDG isosurfaces with different colors according to the electron density multiplied by the sign of the second Hessian eigenvalue using VMD program [71]. From the colored RDG isosurface map and RDG scatter plot of the mixture (L-Pro/2FA DES/GG) as shown in Fig. 8, it could be found that except the hydrogen bond interaction distinguished by blue, Van derWaals forces distinguished by green and steric effect shown by red also coexisted between L-Pro/2FA and GG. AIM was employed to identify the strength of hydrogen bond. The values of ρ_b, $\nabla^2\rho_b$ and hydrogen bond energy at the bond critical point of O_{68}-H_{70}···O_2 and O_3-H_{33}···O_{62} were 0.0393 au, 0.1373 au, −8.026 kcal/mol and 0.0114 au, 0.0432 au, −1.8116 kcal/mol respectively, indicating that both of them were indeed hydrogen bonds and O_{68}-H_{70}···O_2 was much stronger than O_3-H_{33}···O_{62}. This result could be supported by the geometric analysis above.

As the new hydrogen bonds formed between DESs and lignin, changes also occurred to the origin hydrogen bonds in both DESs and lignin. As presented in Table 1, compared to those in L-Pro/2FA, the bond length of two hydrogen bonds between L-Pro and FA in L-Pro/2FA/GG were extended, along with the decrease in hydrogen bond energy, ρ_b and $\nabla^2\rho_b$. For example, the bond length of O_{44}-H_{60} (acidic O—H protons of carboxyl group in L-Pro)···O_{62} (C═O of carboxyl group in FA) in L-Pro/2FA was extended from 1.692 to 1.804 Å, along with the hydrogen bond energy decreased from −9.22 to −6.49 kcal/mol. In addition, O_{68}-H_{70} (acidic O—H protons of carboxyl group in FA)···O_{45} (C═O of carboxyl group in L-Pro) hydrogen bond disappeared in L-Pro/2FA/GG. Even though, the hydrogen bond energy of O_{63}-H_{65}···O_{45} increased slightly from −8.22 to 8.76 kcal/mol. Overall, the hydrogen bond interaction between L-Pro and FA were weakened as L-Pro/2FA interacted with lignin. In regard to GG, the bond length of C_8-O_2 in GG increased from 1.411 to 1.421 Å, as the atom O_2 formed new hydrogen bond with L-Pro/2FA, indicating the structure of GG became loosen which drove the dissolution of lignin in L-Pro/2FA. In this perspective, the microscopic changes including the looser structure of lignin as well as the new formed hydrogen bonds between L-Pro/2FA and lignin, leaded in the macroscopic dissolution behavior of lignin in DESs [72].

According to the analyses above, it was also noteworthy that both the electropositive and electronegative area could form hydrogen bond with GG, which seemed inconsistent with the result we observed in the former section that lignin solubility showed a positive correlation with the value of α (hydrogen-bond donating ability) of DESs but a negative correlation with the value of β (hydrogen-bond accepting ability). This abnormality resulted from the fact that the hydrogen bond O_{68}-H_{70}···O_2 which acidic O—H protons of carboxyl group in FA involved in was much stronger than the one O_3-H_{33}···O_{62} which C═O of carboxyl group in FA involved in, indicating the active proton in -COOH group was more comparative to loosen the structure of lignin. More importantly, the active protons in DESs could be conducive to the lignin degradation into small segments by the cleavage of β-O-4 linkages, further promote the lignin dissolution. The larger mass loss of the regenerated DAL after thermal decomposition test (Fig. 12) indicated that the DAL was partially depolymerized during the dissolution. The obtained smaller

lignin fragments could contact with DESs more substantially, thus enhancing the formation of hydrogen bond between with DESs. The possible route of the cleavage of β-O-4 linkages during lignin dissolution in DESs was elaborated in Fig. 9. After forming hydrogen bond with the active protons in DESs, positive protons are formed at C_α of lignin through acidolysis. Then a resonant stable positive charge was formed through deprotonation and protonation processes, followed by dehydration and further cleavage of the β-O-4 [29,73,74]. Overall, the active protons in DESs played an effective role in the lignin dissolution, whereas the electronegative area showed a negative effect due to the less effective occupation of the active sits in lignin, which hindered the interaction between the active protons and lignin. Therefore, one COOH and one OH group in FA/LA were beneficial to lignin dissolution, by providing active protons to interact with lignin [52]. Whereas, C═O group in LecA resulted in stronger and more complex hydrogen bond network as well as stronger hydrogen bond accepting ability, which hindered the dissolution of lignin.

3.6. DESs recovery and recycling

Besides the dissolution efficacy, the recovery and recycling of DESs have been the essential requirements to accord with sustainable and green technology. Therefore, recyclability experiments of L-Pro/3FA DES were evaluated by conducting the lignin dissolution (60 °C) in L-Pro/3FA for 5 runs, the best overall performing DES among the investigated DESs. Mass balance analysis indicated that 90–95 % of DESs could be recycled in each recycle (Fig. 10). The recovered L-Pro/3FA was used to dissolve DAL at 60 °C. As shown in Fig. 10, a slight decrease in the DAL solubility in L-Pro/3FA was observed with the increasing number of DES recycles, which was ascribed to the impurities such as phenolic compounds derived from the decomposition of lignin during each recycle of lignin dissolution [75]. There is still a plethora of scope left to improve the efficiency of DESs recovery and explore more in-depth understanding on DESs recovery, our study clearly stated that L-Pro/3FA could be recycled and reused in lignin dissolution, which was environmentally friendly and economical.

3.7. Characterization of the original and regenerated DAL

The dissolved DAL in L-Pro/3FA was easily regenerated by adding water as anti-solvent. FTIR spectra, TGA and SEM were used to characterize the origin and regenerated DAL. The origin and regenerated DAL shared the similar FTIR spectra disclosed in Fig. 11, where all the typical peaks of DAL were observed [76], indicating the negligible changes in the backbone structure of the regenerated DAL. Peaks at 1733, 1455, and 1263 were assigned to C═O stretching vibrations, C—H deformations absorption and ring breathing combined with C—O stretching respectively. The peaks at 1585 and 1510 cm^{-1} were attributed to the vibration of the aromatic ring skeleton. 1122 and 1032 cm^{-1} corresponded to the aromatic C—H in-plane deformation of syringyl and guaiacyl, respectively. The thermal decomposition curves and corresponding differential thermal gravimetric analysis (DTGA) from 50 to 700 °C of the origin and regenerated DAL were shown in Fig. 12, and the relevant differences were observed. Both the origin and regenerated lignin showed the fastest mass loss in the temperature range from 250 to 450 °C, due to the major devolatilization reactions [77]. However, after this stage, the regenerated lignin demonstrated a weaker thermal stability (45 wt% mass loss) than the origin lignin (35 wt% mass loss), indicating some changes occurred on the chemical structure of the regenerated DAL. Moreover, the maximum differential thermogravimetric value (DTGmax) of the regenerated DAL was 381 °C, much higher than that (340 °C) of origin DAL. This was because L-Pro/3FA facilitated the proton-catalyzed cleavage of lignin by interacting with eater links, resulting in the generation of lignin molecules with small molecular weight. But on the other hand, the inter/intra C—O and C—C bonds were formed after the cleavage of ether links, which required higher energy to be dissociated [78]. Through the SEM of the origin and regenerated DAL (Fig. 13), the surface of regenerated DAL seemed more rougher and polyporous, while the original DAL showed a more homogeneous architecture. Significantly, small lignin fragments with a cross-linked network structure were observed in the SEM of the regenerated DAL, due to the cleavage of β-O-4 linkages during lignin dissolution in DESs, which arise the possibilities in transformation of lignin into various value-added products [79].

4. Conclusions

In this study, we explored the synergistic effects of different functional groups in DESs on the lignin dissolving ability and further elaborate the lignin solubilization mechanism from the view of molecular interactions by using FTIR, and K–T Solvatochromic Parameters and DFT calculations. The theoretical and spectroscopic investigation revealed that the macroscopic dissolution behavior was due to the newly formed hydrogen bond between lignin and DESs along with the looser structure of lignin. The detailed research on structure-property-lignin dissolving ability of DESs indicated that functional groups of HBD and HBA and their molar ratio showed a vital role on the hydrogen bond interactions within DESs, reflected by the macroscopic property including viscosity and Solvatochromic property, which primarily determined the ability of DESs to form hydrogen bond with lignin. One hydroxyl and one carboxyl group played a positive role in lignin dissolution, due to the ability of providing active proton to form hydrogen bond with lignin. Conversely, the superfluous functional group such as the guanidine group in L-Arg and carbonyl group in LecA leaded to more complex hydrogen bond network and extensive dimer chains, thus resulting in the high viscosity of DESs and low lignin solubility. Moreover, the lignin solubility in the investigated DESs showed a positive correlation with K-T parameters α and π* and a closer positive correlation with the value of α-β, indicating that the donating hydrogen ability played a more important role on lignin dissolution. Among all the investigated DESs, L-Pro/3FA and L-Ala/3FA gave the high DAL solubility of 22.20 and 23.99 wt% at 60 °C, due to the low viscosity and strong net hydrogen-bond donating ability. More importantly, the global ESP maxima and minima of the DESs was positively associated the value of α and β of corresponding DESs, indicating the analysis of ESP quantitative distributions of DESs could provide guidance for designing effective lignin-dissolving DESs or other task-specific DES.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

Yuling Zhang and Hongwei Ren conceived of the presented idea, Yuling Zhang carried out the experiment and wrote the manuscript, Hasmerya Maarof and Syarah Mat Udin supported the QC calculations, Yize Liu, Baochai Li, Jiaoruo Yang contributed the analytical methods,

Dr. Hajar Alias, Prof Erhong Duan and Hongwei Ren supervise the project. All authors discussed the results and contributed to the final manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2023.124829.

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